

Online Loan Eligibility Prediction and Approval Management Platform

A Literature Survey and Comparative Review

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Abstract—Access to formal credit remains constrained by manual, paperwork-heavy loan-approval workflows that are slow, inconsistent, and prone to human bias. The emergence of machine learning, explainable AI, blockchain, and intelligent document processing has motivated a wave of research aimed at automating loan-eligibility assessment and approval decisioning. This paper presents a literature survey of twenty-eight research works spanning classical and ensemble machine-learning models for loan-eligibility prediction, deep-learning and class-imbalance handling techniques for default risk, explainable AI for transparent credit decisions, blockchain and smart-contract-based loan management, alternative-data credit scoring for financial inclusion, intelligent KYC and document-verification automation, and cloud-native microservice architectures for lending platforms. The objectives, methodologies, datasets, and key findings of the surveyed works are summarized and compared using a structured comparison table. Based on this review, the paper identifies the limitations of existing systems and the research gaps that motivate the design of an integrated Online Loan Eligibility Prediction and Approval Management Platform, and outlines future research directions for such a system.

Keywords—*Loan eligibility prediction, loan approval, credit scoring, machine learning, deep learning, explainable AI, blockchain, financial inclusion, KYC automation, literature survey*

I. INTRODUCTION

Access to credit is a cornerstone of individual financial well-being and small-business growth, yet the process of applying for and being approved for a loan remains slow, opaque, and heavily dependent on manual underwriting. Loan officers must verify income and identity documents, cross-check credit history, and apply institution-specific eligibility rules, a process that can take days and is susceptible to inconsistency and unconscious bias. At the same time, banks and non-banking financial companies (NBFCs) face growing pressure to process a rising volume of digital loan applications quickly while still controlling default risk and satisfying regulatory compliance requirements such as Know-Your-Customer (KYC) norms.

An Online Loan Eligibility Prediction and Approval Management Platform aims to address this gap by combining automated, data-driven eligibility scoring with a streamlined digital approval workflow, all within a single application. Before designing such a system, it is essential to understand what has already been attempted in the literature, what technologies and datasets have been used, what results have been achieved, and what limitations remain. This paper therefore conducts a structured literature survey of research relevant to the assigned problem statement, drawn from IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, arXiv, and other scholarly and industry sources.

The remainder of this paper is organized as follows. Section II describes the review methodology. Section III presents the literature survey, grouped into six thematic sub-areas. Section IV presents a comparative analysis of representative papers. Section V discusses the limitations of existing approaches. Section VI identifies the research gaps. Section VII outlines proposed future directions for the Online Loan Eligibility Prediction and Approval Management Platform, and Section VIII concludes the paper.

II. REVIEW METHODOLOGY

Papers and technical reports were identified using keyword searches such as "loan eligibility prediction machine learning," "loan approval prediction," "credit risk ensemble learning," "explainable AI credit scoring," "blockchain loan management," "alternative data credit scoring," "KYC document automation," and "microservices fintech architecture." Sources were selected based on relevance to loan-eligibility or loan-approval decisioning, recency, and the presence of a clearly stated objective, methodology, and outcome. Twenty-eight peer-reviewed papers, preprints, and reputable industry/technical reports were shortlisted and reviewed in depth; their objectives, methods, datasets/tools, and findings were extracted and compared, as reported in Sections III and IV.

III. LITERATURE SURVEY

A. Classical Machine Learning Approaches for Loan Eligibility and Approval Prediction

Early loan-eligibility systems apply supervised classifiers directly to structured applicant data. A machine-learning model built to predict health-loan eligibility from basic applicant-entered values demonstrates a simple, form-driven prediction pipeline suitable for lightweight web applications [1]. Comparative studies evaluating multiple ML models for bank-loan eligibility report that ensemble methods such as Random Forest generally outperform single classifiers on standard applicant datasets [2]. A closely related study applies logistic regression, decision trees, and random forest to loan data, arguing that automated prediction can substantially shorten the loan-sanctioning period for both applicants and bank staff [4]. Reviews synthesizing multiple prior loan-approval studies similarly find that Random Forest consistently achieves the highest accuracy among classical algorithms, while cautioning that algorithmic performance and precision remain an open challenge as new applicant populations and data distributions are encountered [5]. A recent comparative study of Logistic Regression, Random Forest, AdaBoost, and XGBoost on a public bank-loan dataset reports accuracies approaching or exceeding 99% for boosting-based ensembles, and separately notes that Random Forest and XGBoost are effective at predicting mortgage default using macroeconomic predictors such as GDP growth and debt-service-to-income ratio [6].

B. Data Transformation, Feature Engineering, and Credit-Risk Assessment

Beyond model choice, several works show that how data is prepared has a decisive effect on loan-eligibility prediction quality. A novel loan-eligibility prediction (LEP) framework introduces an adaptive Weight-of-Evidence data-transformation method that preserves applicant privacy while improving downstream model performance, following a pipeline of data cleaning, missing-value handling, categorical encoding, and Pearson-correlation-based redundancy reduction [3]. A systematic literature review of machine-learning and metaheuristic optimization techniques for individual credit-risk assessment surveys feature-selection and optimization strategies across the field, concluding that diverse datasets, robust evaluation metrics, and careful feature engineering are essential for model reliability and generalizability [7]. A broader systematic review of machine learning for fraud detection in digital banking, synthesizing evidence from over one hundred peer-reviewed studies under PRISMA guidelines, finds that supervised methods such as decision trees, logistic regression, and support vector machines remain dominant for their interpretability, while unsupervised anomaly detection is increasingly used for novel fraud patterns in highly imbalanced data — a concern directly relevant to loan-fraud and default screening [8].

C. Ensemble Learning and Class-Imbalance Handling for Default Risk

Because default cases are a minority class in most lending datasets, a substantial body of work focuses on resampling and ensemble strategies. A comparative study of Random Forest, SVM, XGBoost, AdaBoost, and multilayer perceptron models on the LendingClub dataset finds that SMOTE combined with Edited Nearest Neighbours (SMOTE+ENN) delivers the best single-technique performance, and that a subsequent voting/stacking ensemble improves accuracy further to 93.7% [9]. A related study using LightGBM and multilayer-perceptron base learners within five ensemble-integration strategies, with SMOTE applied strictly to the training partition to avoid data leakage, reports that anti-leakage preprocessing pipelines are essential for realistic evaluation of loan-default models [10]. Work on imbalanced bank-loan classification similarly combines a Modified-SMOTE resampling step with ensemble classifiers, arguing that neither resampling nor ensembling alone is sufficient and that the two must be combined for robust default prediction [11]. A multi-objective ensemble-learning scheme tailored to credit ratings further demonstrates that combining several sampling techniques with cost-sensitive learning improves default-identification ability under skewed class distributions [12].

D. Explainable AI (XAI) for Transparent Loan Decisions

As ML-based loan decisioning has matured, transparency and regulatory compliance have become central concerns, giving rise to a growing explainable-AI (XAI) literature. A Random-Forest-based loan-approval model enhanced with SHAP and LIME explanations achieves 99% accuracy on a public loan dataset while using feature-importance analysis to clarify which applicant attributes drove each decision, improving trust for both loan officers and applicants [13]. A related model proposed specifically for loan-approval prediction using explainable AI similarly emphasizes that interpretability, not just accuracy, is necessary for institutional adoption [14]. Broader reviews of XAI in fintech credit scoring and risk management describe how SHAP and LIME allow institutions to justify credit-score outcomes to customers and regulators, supporting compliance with fairness and disclosure requirements such as GDPR and the Fair Credit Reporting Act [15]. Complementary work proposes objective metrics for quantifying the degree of explainability of a credit-approval neural network, illustrating a concrete use case in which a bank's network predicts repayment likelihood and an explanation module clarifies why an application with borderline risk was approved or denied [16]. Other studies build interpretable credit-scoring models directly — for example, combining XGBoost with a 360-degree explanation framework offering global, local feature-based, and local instance-based explanations on the HELOC and LendingClub datasets [17], and proposing end-to-end explainable-AI frameworks for credit evaluation that aim to improve consumer trust and reduce false positives in credit decisions [18].

E. Blockchain and Smart-Contract-Based Loan Management

To improve transparency and reduce trust deficits between borrowers and lenders, several papers propose blockchain-based loan and peer-to-peer (P2P) lending architectures. A smart-contract design for P2P lending marketplaces argues that lender distrust in borrower quality, a major friction in existing P2P platforms, can be mitigated by recording borrower and transaction history immutably on a blockchain [19]. A blockchain-infused web platform for smart-contract-enabled loan management describes an end-to-end architecture in which a client's loan request is processed through a smart-contract API, verified by the lender, and settled on-chain, with each step cryptographically secured [20]. A related design proposes a permissioned Hyperledger-Fabric-based lending system that uses digital signatures and oracles to protect applicant data privacy while enabling automated, auditable loan opening and closing rules, demonstrating feasibility on a real-world-inspired lending scenario [21]. Across these works, permissioned or hybrid

blockchain architectures are generally favored over fully public ledgers because they balance transaction transparency with the confidentiality that lending data requires.

F. Alternative Data and Credit Scoring for Financial Inclusion

A parallel research stream addresses loan eligibility for applicants who lack a conventional credit history. Studies on alternative credit scoring show that incorporating psychometric data, social-media activity, and mobile-usage patterns into ML-based scoring models can reduce default rates by roughly nine percent compared with traditional credit models, extending credit access to previously unbanked populations [22]. Research specifically examining non-call-detail-record (non-CDR) mobile-phone data finds that such alternative signals carry meaningful predictive power for credit scoring in emerging-market contexts where formal credit bureau data is sparse or unavailable [23]. A broader evaluation of alternative data sources against a comprehensive credit-scoring dataset similarly concludes that alternative data materially improves scoring accuracy for thin-file and no-file applicants, while noting that model complexity and dimensionality must be carefully managed [24]. Collectively, this body of work suggests that an eligibility-prediction platform intended for broad accessibility should support optional alternative-data inputs alongside conventional financial and demographic fields.

G. Intelligent Document Processing, KYC Automation, and Cloud-Native Architecture

Loan eligibility prediction is only one stage of the end-to-end approval pipeline; document verification and system architecture are equally critical. Industry-oriented studies of intelligent document processing (IDP) describe how combining OCR with NLP and machine learning enables automated extraction, classification, and consistency checking of KYC documents such as identity proofs, salary slips, and bank statements, while flagging document inconsistencies and repayment-behavior anomalies as fraud indicators [25]. A related industry white paper on AI-enabled KYC systems highlights how conversational AI and NLP-driven self-service channels can reduce KYC completion friction by guiding applicants through document submission and clarifying mismatched information in real time [26]. On the architectural side, a comparative study of serverless and microservices architecture patterns in the fintech sector finds that microservices offer lower latency and independent scalability for individual services (such as eligibility scoring, document verification, and notification), while serverless computing reduces operational overhead for bursty workloads, leading many fintech firms to adopt a hybrid of the two [27]. Complementary industry guides on automated loan processing describe how Robotic Process Automation (RPA), OCR, and ML-based risk profiling are typically composed into a single pipeline that spans application intake, KYC verification, credit assessment, and decision-engine-based approval or rejection [28].

IV. COMPARATIVE ANALYSIS

Table I summarizes the focus area, methodology, dataset/tools, key findings, and limitations of eight representative papers spanning the thematic areas discussed in Section III.

TABLE I
Comparative Analysis of Representative Reviewed Papers

Ref.	Focus Area	Methodology / Technology	Dataset / Tools	Key Findings	Limitations
[3] Sana & Rahman (2025)	Loan eligibility prediction (LEP)	Adaptive Weight-of-Evidence data transformation + ML pipeline	Bank customer loan dataset	Privacy-preserving transformation improves LEP accuracy over raw-feature baselines	Single-institution data; generalizability untested
[6] Haque (2024)	Bank loan approval prediction	Logistic Regression, Random Forest, AdaBoost, XGBoost	Kaggle bank loan dataset	AdaBoost reaches up to 99.99% accuracy; ensembles outperform single classifiers	Very high accuracy suggests possible overfitting / data leakage risk
[9] Sheffield Hallam (2024)	Loan default risk prediction	RF, SVM, XGBoost, AdaBoost, MLP + SMOTE/ENN resampling, voting/stacking	LendingClub dataset	SMOTE+ENN best single technique (90.5% acc.); stacked ensemble reaches 93.7% accuracy	Performance gains depend heavily on resampling choice; class-imbalance sensitivity
[13] Ravi et al. (2026)	Explainable loan approval	Random Forest + SHAP/LIME explanations	Kaggle loan dataset	99% accuracy with RF; XAI clarifies key predictors, improving trust	Explanation fidelity vs. model complexity trade-off not fully resolved
[19] IEEE (2023)	P2P lending trust	Blockchain smart contracts for borrower selection	Field study / prototype	Blockchain improves lender control and transparency in marketplace lending	Consensus/throughput overhead; regulatory uncertainty for DeFi-style lending
[21] IEEE (2024)	Blockchain loan management	Hyperledger Fabric smart contracts, digital signatures, oracles	Simulated lending scenario	Improves data-privacy and auditability of loan disbursement and closure	Scalability of permissioned blockchain under high transaction volume
[22] ResearchGate (2025)	Financial-inclusion credit scoring	ML on psychometric, social-media and mobile-usage data	Alternative-data corpus	Alternative data reduces default rates by ~9% vs. traditional models	Data-privacy and consent concerns; model bias across demographics
[25] Auxiliobits (2025)	KYC & document automation	OCR + NLP intelligent document processing, rule engines	Applicant-submitted ID/financial documents	Automates document parsing, consistency checks and fraud-flagging in KYC	Struggles with damaged/non-standard documents; needs human-in-the-loop review

V. LIMITATIONS OF EXISTING APPROACHES

Across the reviewed literature, several recurring limitations emerge. First, most loan-eligibility ML models are trained and evaluated on small, single-institution, or publicly released benchmark datasets (frequently the same Kaggle or LendingClub datasets), raising concerns about generalizability to new geographies, applicant populations, and loan products. Second, several studies report extremely high accuracies (in some cases approaching 100%), which is more likely to indicate data leakage, small hold-out samples, or narrowly defined applicant populations than genuine real-world performance, and few papers report calibration or fairness metrics alongside accuracy. Third, explainable-AI techniques such as SHAP and LIME are typically demonstrated as standalone add-ons to a trained model rather than integrated into a production approval workflow, so it remains unclear how explanations should be surfaced to applicants and

loan officers in practice. Fourth, blockchain-based lending architectures are largely validated through simulations or small prototypes rather than production-scale deployments, leaving open questions about consensus throughput and regulatory acceptance. Finally, alternative-data credit-scoring approaches raise unresolved privacy, consent, and algorithmic-bias concerns that are acknowledged but not fully addressed in the surveyed works [8], [15], [22], [24].

VI. RESEARCH GAPS

Based on the above analysis, the following research gaps are identified:

- Lack of a single, integrated platform that combines eligibility scoring, explainable decisioning, KYC/document verification, and approval-workflow management within one system, rather than as separate research prototypes.
- Sparse evaluation of loan-eligibility ML models on diverse, large-scale, real-world (rather than small or single-source) applicant datasets, limiting confidence in generalizability.
- Insufficient integration of explainability techniques (SHAP/LIME) into end-to-end approval workflows that loan officers and applicants can actually interact with, rather than offline model-analysis exercises.
- Limited exploration of privacy-preserving methods (e.g., federated learning, on-device scoring) for incorporating alternative data into eligibility prediction without exposing sensitive applicant behavior.
- Underexplored combination of blockchain-based audit trails with ML-based eligibility scoring in a single, production-representative architecture.
- Few studies report fairness or bias metrics across demographic subgroups for loan-approval models, despite this being a well-known regulatory and ethical concern.

VII. PROPOSED FUTURE DIRECTIONS

Building on the identified gaps, an Online Loan Eligibility Prediction and Approval Management Platform should combine (i) an ensemble-based eligibility-prediction engine informed by the classical and ensemble ML techniques surveyed in Sections III-A to III-C [2], [6], [9], [12]; (ii) an explainability layer using SHAP/LIME-style feature attribution surfaced directly within the loan-officer and applicant interfaces, in the spirit of the XAI approaches reviewed in Section III-D [13], [15], [17]; (iii) an optional alternative-data module for applicants with thin or no formal credit history, following the financial-inclusion research reviewed in Section III-F [22], [23]; (iv) an intelligent document-processing and KYC-automation layer combining OCR and NLP to verify identity and income documents, drawing on the approaches surveyed in Section III-G [25], [26]; (v) an optional blockchain-backed audit trail for recording key approval decisions and disbursement events immutably, following the smart-contract designs reviewed in Section III-E [20], [21]; and (vi) a cloud-native, microservices-based backend so that eligibility scoring, document verification, and notification/approval-workflow services can each be developed, scaled, and secured independently [27], [28]. Future empirical work should validate such an integrated system on large-scale, demographically diverse applicant data, and should report fairness and explainability metrics alongside conventional accuracy, precision, and recall.

VIII. CONCLUSION

This paper surveyed twenty-eight research works relevant to the design of an Online Loan Eligibility Prediction and Approval Management Platform, covering classical and ensemble machine learning for loan-eligibility and default prediction, class-imbalance handling techniques, explainable AI for transparent credit decisions, blockchain-based loan management, alternative-data credit scoring for financial inclusion, and intelligent KYC/document-automation and cloud-native architectures for lending platforms. While substantial progress has been made in each individual sub-area, the literature reveals a clear gap: no reviewed system integrates accurate, explainable eligibility prediction with automated KYC verification and an auditable, scalable approval workflow in a single platform. Addressing this gap, as outlined in Section VII, represents a concrete and well-motivated direction for the proposed system.

REFERENCES

- [1] "Loan Eligibility Prediction using Machine Learning based on Personal Information," in Proc. IEEE Conf., IEEE Xplore, 2023.
- [2] "Machine Learning Models for Predicting Bank Loan Eligibility," ResearchGate preprint, 2022.
- [3] J. K. Sana and M. S. Rahman, "A Novel Loan Eligibility Prediction Model with Effective Use of Data Transformation Methods," Expert Systems with Applications, ScienceDirect, 2025.
- [4] "An Approach to Predict Loan Eligibility using Machine Learning," in Proc. Int. Conf. Artificial Intelligence and Data Engineering (AIDE), IEEE, 2022.
- [5] Tejaswini et al., "Predicting Bank Loan Eligibility Using Machine Learning," in Proc. IEOM Society Conf., 2022 (citing Srivastava et al. 2018; Karthiban, Ambika, and Kannammal 2019).
- [6] F. M. A. Haque, "Bank Loan Prediction Using Machine Learning Techniques," arXiv:2410.08886, 2024 (citing Krasovytskyi 2023; Muhammad Iqbal 2023).
- [7] "Machine Learning and Metaheuristics Approach for Individual Credit Risk Assessment: A Systematic Literature Review," PMC, 2025.
- [8] "Machine Learning for Fraud Detection in Digital Banking: A Systematic Literature Review," arXiv:2510.05167, 2025.
- [9] "Ensemble-Based Machine Learning Algorithm for Loan Default Risk Prediction," Sheffield Hallam University Research Archive, 2024.
- [10] "Research on Loan Default Prediction Methods Based on Ensemble Learning," in Proc. Int. Conf. Big Data, Information and Computer Network (BDICN), ACM, 2026.
- [11] "Bank Loan Classification of Imbalanced Dataset Using Machine Learning Approach," Research Square preprint, 2023.
- [12] "Loan Default Prediction Using a Credit Rating-Specific and Multi-Objective Ensemble Learning Scheme," Expert Systems with Applications, ScienceDirect, 2023.
- [13] V. Ravi et al., "Explainable AI (XAI) for Credit Scoring and Loan Approvals," in Proc. Int. Conf. Web 6.0 and Industry 6.0 (WIN), Lecture Notes in Networks and Systems, Springer, 2026.
- [14] O. M. Ayad, A.-E. F. Hegazy, and A. Dahroug, "A Proposed Model for Loan Approval Prediction Using Explainable Artificial Intelligence," in Proc. Int. Conf. Intelligent Computing and Information Systems (ICICIS), IEEE, 2023.
- [15] "Harnessing Explainable AI (XAI) for Transparency in Credit Scoring and Risk Management in Fintech," ResearchGate preprint, 2025.